



Recycled filtered contaminants from liquid-fed pyrolysis as novel building composite material

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ABSTRACT

Liquid-fed pyrolysis allows the conversion of contaminated postconsumer plastic waste into valuable resources, removing contaminants through wax dissolution and filtration. One of the main challenges is currently represented by the management of its main byproduct, the contaminant-rich retentate from the filtration process. New circular economy strategies are needed to use this waste plastic-based composite as secondary raw materials. Despite the increasing trend in using plastic and plastic-waste composites for the building sector, there are currently limited applications of industrial recycling waste as engineering construction materials, e.g., from pyrolysis. This study evaluates the suitability of contaminant-rich retentate from liquid-fed pyrolysis of postconsumer multilayer packaging waste as novel composite materials for the building sector, taking advantage of its intrinsic composite nature. Thermogravimetric analysis was conducted to assess the composition of the retentate from filtration, a mixture of wax, polyethylene, polyethylene terephthalate, and aluminum. The compressive mechanical properties and densities were then evaluated on samples obtained through hot compression molding, using two batches to assess possible anisotropic behavior from the manufacturing process or the aluminum part orientation. The results indicate the suitability of the waste composite material for the building sector, reaching compression strengths (10–12 MPa) superior to construction bricks and brickworks (7 MPa), as well as 57 % lower density, 0.77 g/cm³. The ductile fracture behavior indicates its potential use for applications requiring failure prediction and safety constraints. This high strength-to-weight ratio composite represents a valuable alternative to virgin materials, showing potential for structural and aesthetic applications, including lightweight bricks, interior textured panels, decorative facades, and customized pavement tiles and slabs. Hot compression molding of pyrolysis waste composite paves the way for the real use of plastic-based composites from mixed and contaminated industrial waste in the building sector, contrasting resource depletion.

1. Introduction

Recent advancements in sustainable material technologies, e.g., green nanotechnology, recycling methods, and novel composite

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formulations, offer potential solutions to the environmental challenges faced over the last several decades [1]. These technologies aim to address current issues in sustainable production and consumption practices, such as resource depletion, environmental remediation, renewable energy production, and global waste reduction. Their use spans several application sectors, ranging from wastewater treatment [2,3] to the mitigation of greenhouse gas (GHG) emissions [4], as well as in the oil and gas industry [5]. Considering the significant impact of the construction sector, novel technologies seek to find new alternative materials to reduce the GHG emissions of currently available building materials. This urgency is because the CO₂ emissions from the cement industry reach ~7 % of the global emissions [6,7], prioritizing the development of new alternatives in this sector. To this end, novel material options have been studied, including geopolymers [4,8], composites filled with recycling construction and brick debris [4,9], novel binders from abundant resources [10–12], and recycled materials from existing waste streams, e.g., plastic waste [13], or plastics and sand [14–16] fostering waste reduction and contrasting resource depletion.

Pyrolysis is a promising polymer recycling technology to enable circular economy approaches based on material recovery [17]. One of the primary challenges facing current pyrolysis technologies is the presence of contaminants in plastic waste streams [18]. Most plastic waste, especially from consumer sources, contains some level of contamination, which adversely affects both the operation of the pyrolysis process and the quality of the pyrolysis products [18]. Furthermore, there are still open issues in using pyrolysis technologies with mixed plastic waste streams, e.g., containing different disposed products and composite packaging [17]. Although they represent more realistic and common waste compositions, a mix of different polymers or other engineering materials is more difficult to treat with these technologies, as well as to achieve optimal pyrolysis conditions [18]. Liquid-fed pyrolysis, a method that couples dissolution with pyrolysis to convert plastic waste, seeks to mitigate contamination issues by preprocessing the plastic waste before pyrolysis, i.e., through dissolution. By selectively dissolving polyolefin plastics in pyrolysis wax solvent [19], insoluble contaminants such as metals, polyethylene terephthalate (PET), paper, and cardboard are removed from the feed through liquid filtration. This pre-treatment facilitates the collection of different products, such as gas for power generation, liquids for bioconversion feedstock, and wax products to be reused for new liquid filtrations [19,20], closing the loop and promoting process circularity [20]. This new step, liquid filtration, solves the pyrolysis contamination issue but creates a contaminant-rich retentate at the end of the filtration process. Due to the viscous nature of dissolved polyethylene (PE) in wax, a significant amount of wax and plastic clings to the filtered contaminants [21], creating a mixture of plastics and metals to be handled at the end of the process.

To this end, this contaminant-rich retentate represents a potential secondary raw material to replace virgin resources, aiming for cleaner and circular production models [22]. Understanding the properties through characterization techniques can help detect alternative uses of this byproduct, a waste plastic-based composite, matching it with potential application sectors [21]. Researchers are currently investigating the use of recycled plastics and plastic-based waste as replacements for virgin materials, such as thermoplastics [23,24], reinforced concrete [8,25], or fiber-reinforced polymers [26–28], including natural fibers [29]. First, their use can reduce the need for pristine materials. Thermomechanical degradation of plastic-based waste, however, can significantly affect the performances of the manufactured parts, e.g., leading to downcycling or requiring a blend with virgin materials [30]. This interest is further nurtured by the significant amount of waste plastic produced globally, which is expected to exceed 500 million metric tons over the next decade [31], requiring new scalable and more economically viable recycling solutions. Despite the current issues in managing waste plastic and plastic composite disposal [30], finding new uses can reduce their environmental impact and create new market opportunities by keeping them in use through time for value-added applications. From this perspective, composite waste materials have become an interesting option thanks to their heterogeneous composition, leading to unique properties that cannot be achieved with pure materials, e.g., mechanical properties and high strength-to-weight ratios. These properties make composite waste material suitable for the construction and building sectors, a relatively new application sector for similar secondary raw materials [13]. In the last few years, they have been increasingly studied as potential construction materials, exploiting them in several ways. For instance, composite waste materials can be used as aggregates for concrete or brick, e.g., for masonry structures and outdoor applications [14,15]; fiber reinforcements and fillers, e.g., road construction [32] and reinforced mortars [33,34]; and structure preparation, e.g., cobbled roads or panel boards [16,35]. This choice is mainly due to their potential to reduce resource depletion, such as cement and clay consumption as non-renewable resources, and limit the environmental impacts connected to conventional construction material manufacturing, e.g., high-temperature firing, as well as the GHG emissions involved in brick production [15]. Moreover, using composite waste materials fosters distributed recycling waste streams in new sectors, increases plastic waste value through recycling, and exploits new property combinations only achievable using composite materials [13]. Currently, their widespread adoption is still limited by several challenges, such as variability and inconsistencies between waste batches [16,30], long-term durability in different weather conditions [13,16], and actual costs in their implementation [13,15], as well as missing standards and regulations for their use and future disposal [13,30]. Nevertheless, their use can stimulate the study of new real-world applications with considerable environmental promise [36], which are still limited despite the growing number of works studying their properties.

In this context, the use of pyrolysis waste as construction materials represents a potential way to reduce resource depletion in the building sector, providing novel combinations of material properties. Currently, pyrolysis byproducts for construction have been scarcely investigated, although similar byproducts contain material waste suitable for the building sector, such as PE and PET [13,21]. From the literature, pyrolysis waste is mainly explored as a source for fuel oils [37], potential additives for bitumen [38], or asphalt binders [10]. According to existing research, pyrolysis waste has not been exploited as the main composite material of construction parts and structures, considering them waste to be handled rather than new secondary raw materials. To this end, studying their properties can pave the way to new applications, avoiding their disposal and component sorting. Exploiting their heterogeneous composition as composite materials represents a novel approach to repurposing these byproducts. Moreover, it helps reduce the reprocessing steps needed for their use as secondary raw materials and, at the same time, provides new material property combinations to the construction market.

This work explores the suitability of contaminant-rich retentate from liquid-fed pyrolysis of multilayer packaging waste, i.e., military meals-ready-to-eat (MRE) packaging, as an alternative composite material for construction. The main aim is to use a real byproduct from recycling processes as a secondary raw material for novel applications, taking advantage of the specific composition of the MRE-based retentate and offering properties not achievable with traditional construction materials. After filtering the MRE-based retentate from the pyrolysis process, the material was analyzed by means of thermogravimetric analysis (TGA), ensuring the successful removal of contaminants from the pyrolysis process and defining the composition of this novel composite (Subsection 3.1). Some samples were then manufactured through hot compression molding by using a low-cost hot press to qualitatively assess the manufacturability of the material (Subsection 3.2), as well as evaluate the compressive mechanical properties and average density (Subsection 3.3). The experimentation aims to validate the suitability of this composite waste material for the building sector, detecting potential applications and comparing its use with conventional materials and products for construction (Subsection 3.4). Finally, possible limitations and future research work are highlighted to provide the foundation for the use of pyrolysis byproducts as construction materials (Subsection 3.5). The novelty of this work lies in the repurposing of pyrolysis waste as a resource for new composite materials with unique properties, overcoming the challenges in handling heterogeneous waste by using it for applications with high resource depletion ratings.

2. Materials and methods

2.1. Building material from industrial recycling byproducts

This work investigates the use of byproducts from a novel pyrolysis process as potential materials for the building sector. The tested building material entirely derives from the liquid-fed pyrolysis process of multilayer food packaging, the MRE packaging, representing an abundantly available waste source made of different materials. MREs contain two types of packaging: pure (3 wt% nano clay-filled) PE bags and multilayer food pouches. The multilayer packaging consists of three layers: PE (no filler) (62.5 wt%), aluminum (24.2 wt%), and PET (13.3 wt%). Overall, this model input waste stream is composed of 87.4 wt% PE, 8.1 wt% aluminum, and 4.5 wt% PET. MREs are currently difficult to recycle with conventional recycling processes due to their multilayer composition and the different material types involved. As explained in Subsection 2.2, the liquid-fed pyrolysis process allows the recovery of different engineering materials that are otherwise difficult to separate, including multilayer packaging and MREs. It still generates byproduct waste to be properly handled, however, represented by the retentate residues from the filtration step. This retentate was, therefore, selected to be tested as a new potential fully recycled building material, using this byproduct as the starting point of this work.

After the MRE-based retentate collection from the filter (Subsection 2.2), the material was used as it is, avoiding further modification of its original composition and further mix design, minimizing its reprocessing steps. Accordingly, the building material is fully made of the MRE-based retentate, whose mixture composition depends on the materials of the recycled MREs and the collected contaminants. In this case, the collected MRE-based retentate consisted of a mixture of pyrolysis wax, PE, PET, and aluminum particles, and the mixture tested in this work corresponds to 100 wt% MRE-based retentate. The specific percentages of this mixture composition required to be assessed after the retentate mixture collection. To this end, samples of the MRE-based retentate were tested using TGA, detecting its composition after the liquid-fed pyrolysis. The tests were conducted with a model TGA701 from LECO Corporation (St Joseph, MI, US). One-gram samples were loaded into ceramic crucibles and heated at a rate of 1 °C/min until they reached 850 °C under a constant purge of nitrogen. Samples were tested in duplicate to ensure instrument accuracy. The specific mixture composition is presented in Subsection 3.1.

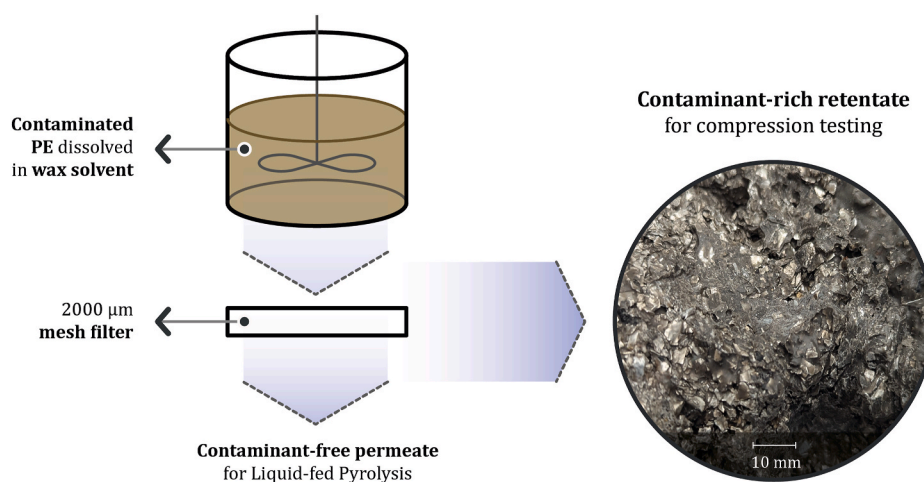


Fig. 1. Contaminated polyethylene (PE) was dissolved in pyrolysis wax solvent and passed through a 10-mesh filter. The insoluble contaminants (mainly aluminum) were concentrated in the MRE-based retentate.

2.2. Collection of the MRE-based retentate from liquid-fed pyrolysis

Liquid-fed pyrolysis is a novel recycling process to recover materials and components from plastic waste starting from their dissolved liquid feed obtained from a pyrolysis wax solvent [20]. This approach allows gas collection for power generation, liquids for upcycling, and wax products to be recycled in the same pyrolysis process [19,20].

For this work, a complex waste plastics stream was created based on military MRE packaging. MREs were selected as the main recycling waste source, considering their average annual waste generation amounts in tons [39]. The liquid-fed pyrolysis process was carried out with an open-source controller-based pyrolysis system consisting of a dissolution tank, a reactor for pyrolysis, and a set of condensers [40]. The waste stream was initially shredded using a 3Devo hybrid shredder (Utrecht, Netherlands) to a particle size range of 2000–6500 μm [41]. As shown in Fig. 1, the shredded plastic/aluminum mixture is then dissolved in a 1:1 ratio of pyrolysis wax at 240 °C. Once dissolved, the liquid-feed underwent dead-end filtration using a 2000 μm (10 U.S.) mesh filter under a pressure of 34, 473.8 Pa (5 Psig). A scraper was then used to continuously remove filtered contaminants off the screen to prevent clogging. The permeate that passes through the filter has a reduced contaminant level and is pyrolyzed to create wax, liquid, and gas hydrocarbon products [20]. Filtration continued until the flow of the permeate ceased or could no longer be achieved. The retentate from the MRE recycling, which means the byproduct material tested in this work, was then removed from the filter screen and prepared for compression testing, i.e., hot pressing of the mechanical samples.

2.3. Sample production through hot pressing

The MRE-based retentate collected from liquid-fed pyrolysis was crushed to reach an average dimension of 10 mm and then used to fabricate prism samples for compression tests without adding other components or water. Hot pressing, or hot compression molding, was selected as the main manufacturing process for the experimentation, using the custom low-cost hot press apparatus shown in Fig. 2a [42]. This process represents a suitable way to process plastic-based composites and manufacture bricks or construction elements made of recycled or composite materials [16,43], improving the compression ratio and the overall performance of the final parts. Contrarily to conventional brick production, plastic-based composites do not require any drying or firing steps, reducing manufacturing times, optimizing resource and energy usage, and minimizing shrinkage rates in final parts [43,44].

The samples obtained through hot pressing were also used to qualitatively assess the suitability of the MRE-based retentate to be shaped in semi-finished products, such as bricks. Two batches of prism specimens were manufactured by using a custom compression hot press with a maximum pressure of 0.38 MPa [42] equipped with a mold designed according to the ASTM D695-23 standard [16, 45]. The samples have a nominal dimension of 12.7 mm in width, 12.7 mm in height, and 25.4 mm in length.

The hot press was first preheated at 250 °C for 30 min. The samples were produced using a processing temperature of 200 °C for 20 min to allow the PE/PET/wax matrix to fill the cavities without significant matrix loss from excessive flash. The mold was initially loaded with the material and kept in the press for 5 min to soften the matrix, reducing the presence of internal voids. After this step, the mold was then removed, opened, and filled with additional material to ensure the complete filling of the mold cavities. Some rectangular inserts were added to the cavities to ensure homogeneous compression loads on the material, and the mold was finally closed and pressed for the remaining time. After cooling down the mold and releasing the pressure provided to the mold, the specimens were removed and sanded to eliminate any flash material from the hot pressing (Fig. 2b).

2.4. Compression test

Compression tests were performed on two batches of at least five prism specimens each, manufactured through hot pressing (Subsection 2.3), following the ASTM D695-23 standard [45]. Tests were carried out with a compression testing machine from MTS System Corporation (Edein Prairie, MN, US) equipped with a 250 kN load cell at 1.3 mm/min speed until failure. The actual measures of the samples were checked with a caliper, and the samples were weighted to calculate the average density (ρ) and its standard error.

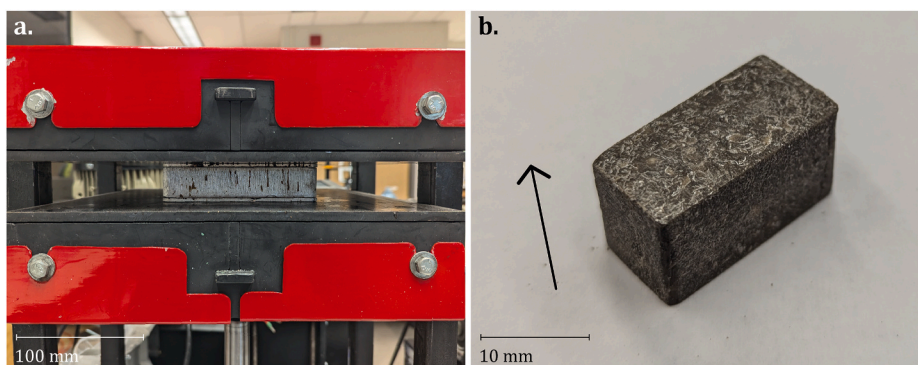


Fig. 2. Sample manufacturing method: (a) low-cost hot press layout during sample compression and (b) shape and dimensions of the samples manufactured with the hot press, highlighting the mold opening direction (indicated by the black arrow).

Two batches of samples were tested to assess a possible anisotropic behavior in the samples due to the manufacturing process or the orientation of the aluminum particles, as shown in Fig. 3. In detail, two different orientations of the manufactured samples were selected during the tests, representing the only difference between the two batches. The first batch of samples (Fig. 3a) was tested by applying the compression load perpendicular to the mold opening direction (Pb), whereas the second batch (Fig. 3b) was positioned by keeping the mold opening direction parallel to the compression load (Pw). The compressive stress-strain curves from the experimental tests were used to calculate the mean values and standard errors of the compressive strength (σ_c), compressive yield strength at 0.2 % offset (σ_{ys}), and modulus of elasticity in compression (E_c).

3. Results and discussion

3.1. MRE-based retentate composition from TGA

The collected retentate is obtained from the liquid-fed pyrolysis of post-consumer multilayer food packaging, the MREs. It currently represents a byproduct of this recycling process without a proper application. As mentioned in Subsection 2.2., postconsumer MREs are made of a mixture of PE, PET, and aluminum layers, making the contaminant-rich retentate heterogeneous composition potentially suitable for new application fields, including the construction sector.

Based on the TGA results conducted on the MRE-based retentate (Fig. 4), the MRE-based retentate is composed of 22.8 ± 3.1 wt% pyrolysis wax, 65.3 ± 3.4 wt% plastic (PE and PET), and 11.8 ± 0.2 wt% aluminum. A mass balance on the filtration found that 84 % of the aluminum was filtered, representing a successful removal of contaminants for pyrolysis. The filtration rate of PET was not able to be determined via TGA as the peaks for PET and PE overlap between 350 and 450 °C. According to the results, this retentate is a plastic-based composite material made of a PE/PET matrix, where aluminum is considered the main dispersed phase, as visible in Fig. 2a. The high PE/PET plastic content ensures a lightweight material, whereas the aluminum particles can improve the mechanical properties of the component. The MRE-based retentate can be used as a plastic-based composite material for structural applications aiming at weight reduction.

Table 1 shows the mixture composition used for the tests and based on the TGA analysis, showing the possible function of each detected material. i.e., binder, aggregate, and reinforcement. The mixture is only composed of the MRE-based retentate, which inherently represents a specific mixture being a composite material. No additional components or water were added to the MRE-based retentate, avoiding any further mixing phase since it can be easily reprocessed through hot pressing. For these reasons, the collected liquid-fed pyrolysis retentate can be used as a potential secondary raw material for new applications, e.g., composite material for the building sector. The PE/PET and wax mixture could be exploited to improve water resistance and heat barrier properties of construction elements, as suggested by the use of wax and PET in previous thermal insulation applications [46–48]. Furthermore, these results make this material composition a potential alternative for energy conservation, either during processing or for the final parts [29,38], requiring further work for a comprehensive thermal characterization and flammability assessment of this composite.

3.2. MRE-based retentate manufacturability

The samples were successfully manufactured by hot pressing, demonstrating the material's manufacturability with conventional manufacturing processes. The compression-molded samples exhibited no visible voids, defects, or changes in color due to thermal degradation during the manufacturing process, showing a texture appearance connected to the heterogeneous material composition (Fig. 2b). This result indicates good homogeneity of the samples, making the specific material texture a potential aesthetic characteristic to be exploited in the final parts. Moreover, the parts were easily removed from the mold without breaking, ensuring part integrity and demonstrating the uniform filling of the mold cavities during the process. The shaped material can also be postprocessed

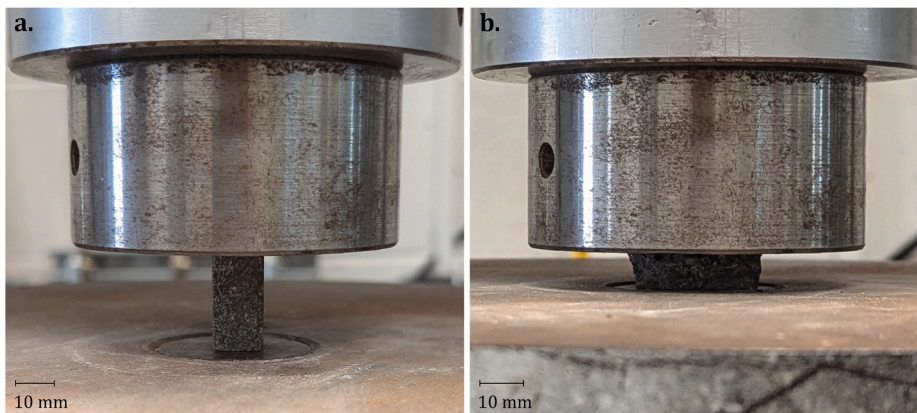


Fig. 3. Sample orientation during the compression tests: (a) first batch test “Pb” (applying the compression load perpendicular to the mold opening direction); and (b) second batch test “Pw” (applying the compression load parallel to the mold opening direction).

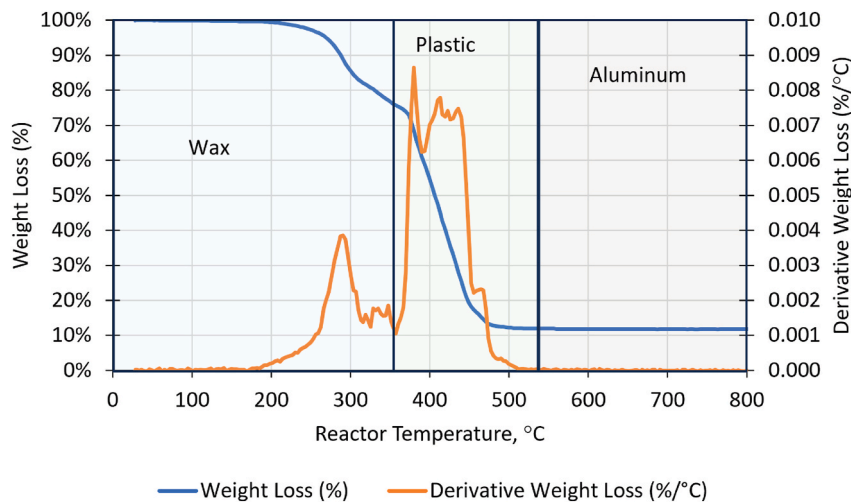


Fig. 4. Composition of the filter MRE-based retentate from liquid-fed pyrolysis found via TGA, corresponding to the mixture composition of the building material: 22.8 ± 3.1 wt% pyrolysis wax, 65.3 ± 3.4 wt% plastic (PE and PET), and 11.8 ± 0.2 wt% aluminum.

Table 1

Mixture composition of the building material tested in this study (assessed through TGA), highlighting the origin of the materials found in the composition, their possible function in the mixture as a building material, weight content, and average density.

Mixture design (building material)	Mixture component function ^b	Material	Origin	Content (wt %)	Source
100 wt% MRE-based retentate (mixture) ^a	Binder	Pyrolysis wax	Liquid-fed pyrolysis process	22.8 ± 3.1	TGA
	Aggregate	PE/PET mixture	MRE (multilayer packaging and bags)	65.3 ± 3.4	TGA
	Reinforcement	Aluminum particles	MREs (multilayer packaging)	11.8 ± 0.2	TGA

^a according to the collected retentate after pyrolysis. No further components and water were added to the mixture.

^b according to the material and content found via TGA, evaluating the specific function of the retrieved materials in the mixture.

with conventional hand tooling or sanded without visible damage to the parts, making it suitable for the potential application sector, e. g., for brick structures. Despite the simple shape of the samples, their ease of moldability suggests a potential use to manufacture more complex parts, for instance, with intricate surface details. Hot-pressed parts for construction can reduce the timings and shrinkages from the drying and firing processes, as well as their energy consumption, overcoming some drawbacks of clay and concrete building elements [43,44]. This capability can further enlarge the possible uses of this byproduct for architectural and decorative elements in construction, taking advantage of the hot pressing process to make small batches of building elements with complex geometries and aesthetic textures.

3.3. Mechanical compressive properties

The compressive properties of the MRE-based retentate were then studied to understand its suitability for structural applications as a replacement of concrete materials, e.g., for construction bricks from hot compression molding. As a matter of fact, hot-pressing bricks can reach better compression ratios, hence improving the compressive strength of the final parts compared to conventional brick manufacturing methods [43,44].

According to the stress-strain curves in Fig. 5, the tested samples exhibited ductile behaviors at compression with some differences according to the test direction, especially in the failure mechanism. According to Fig. 5a and b, the first batch showed higher strains and a shear fracture after a starting ductile behavior, as visible from the deformations on the side walls of the samples. On the contrary, the second batch reached higher stress at fracture with a tendency to a ductile fracture and minor cracks, leading to the progressive collapse of the samples (Fig. 5c and d). The behavior exhibited by the MRE-based retentate differs from the conventional materials used for bricks, such as concrete, which usually show brittle failures and cracks. The MRE-based retentate can be, therefore, exploited in applications where failure predictability is crucial, increasing the safety factor of structures made with this composite formulation.

The compressive properties from the experimental tests are reported in Fig. 6. In general, the results achieved good accuracy and repeatability, showing low variability between the tested samples, i.e., relative standard deviations between 5 and 10 % in most cases. In addition, the specimens show a good consistency in their density (ρ), i.e., 0.77 g/cm^3 , with a relative variation of 6.6 % (Table 2). The results are in line with the high wax and PE content from the starting MREs composition, as well as the retentate composition from

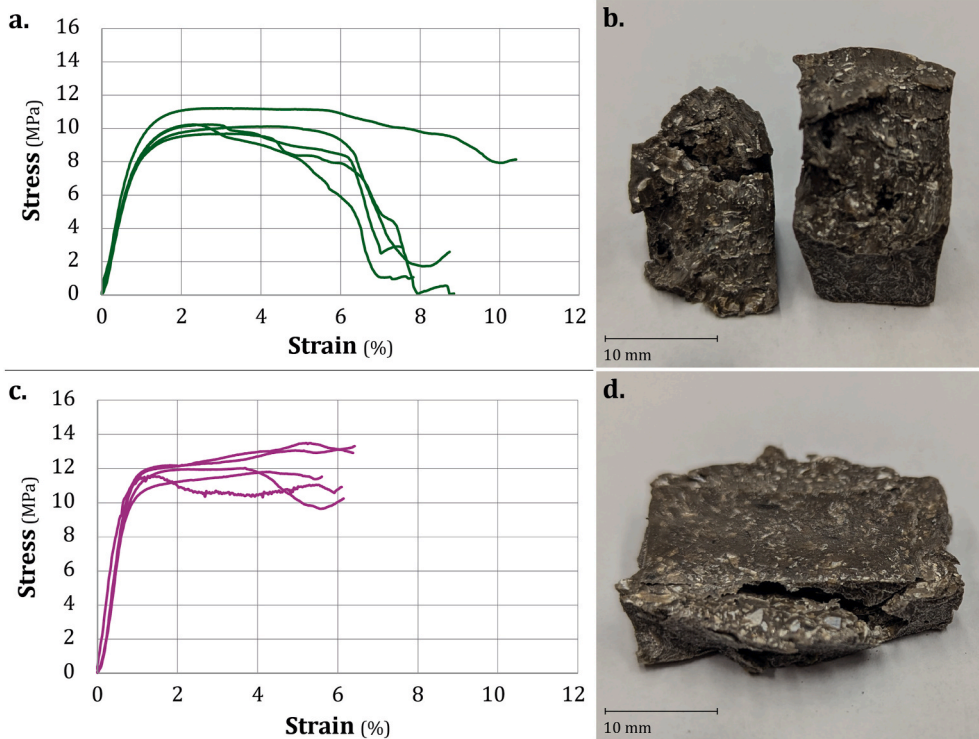


Fig. 5. Results from the compression tests: (a) stress-strain curves of the first batch of samples Pb (applying the compression load perpendicular to the mold opening direction) and (b) a representative specimen after the test showing the failure mode and behavior of batch Pb; (c) stress-strain curves of the second batch of samples Pw (applying the compression load parallel to the mold opening direction) and (d) a representative specimen after the test showing the failure mode and behavior of batch Pw.

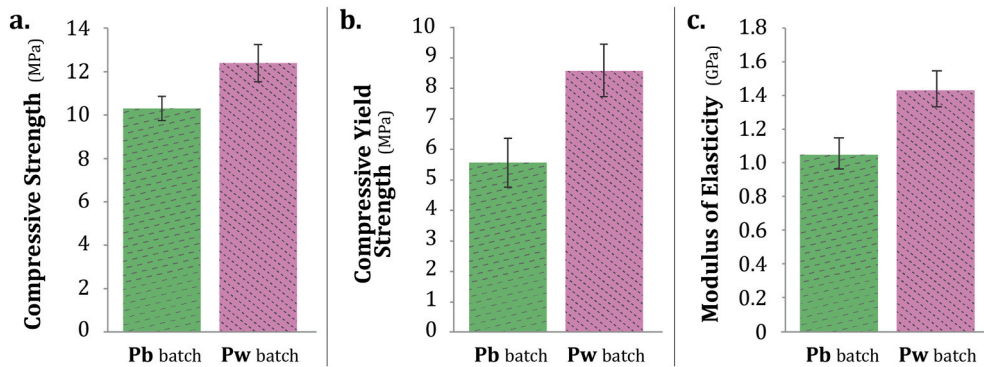


Fig. 6. Results from the compression tests of batch Pb (on the left side of the graphs) and batch Pw (on the right side of the graphs): (a) compressive strength (σ_c); (b) compressive yield strength (σ_{ys}); and (c) modulus of elasticity in compression (E_c).

the TGA (Fig. 4 and Table 1). The value also indicates a possible presence of porosity or voids, being lower than the average densities of wax and PE, $\sim 0.9 \text{ g/cm}^3$, despite the presence of PET and aluminum particles, which have higher density values, i.e., $\sim 1.3 \text{ g/cm}^3$ and 2.7 g/cm^3 , respectively [51].

According to the data, some differences can be noticed by changing the test orientation of the sample, showing an anisotropic behavior of the specimens, and achieving higher stress when the compression load is parallel to the mold opening direction (batch Pw). For instance, the compressive yield strength increases by applying a compression load parallel to the mold opening, i.e., passing from 5.57 to 8.59 MPa. Similar trends are also visible for compressive strength and the modulus of elasticity. This fact indicates a possible anisotropic behavior of the material, although further analyses are needed to confirm this result, such as morphological analysis through optical microscopy or scanning electron microscopy. This behavior can be influenced by the mold opening direction of the samples and the metal particle orientation during their fabrication. The different failure mechanisms can also be due to the different

Table 2

Comparison between the compressive properties achieved by the MRE-based retentate and the main works from the state-of-the-art on composite waste plastic-based materials and recycled plastics for the construction sector, together with a benchmark average of conventional concrete bricks.

Reference	Material	Density ρ (g/cm ³)	Compressive strength σ_c (MPa)	Compressive yield strength σ_{ys} (MPa)	Modulus of elasticity E_c (GPa)	Fracture behavior
This work (Pb)	Mixture of PE, PET, wax, and aluminum from pyrolysis	0.77 ± 0.05 (6.6 %)	10.30 ± 0.56 (5.4 %)	5.57 ± 0.80 (14.4 %)	1.06 ± 0.09 (8.8 %)	Ductile, shear
This work (Pw)			12.39 ± 0.86 (6.9 %)	8.59 ± 0.85 (9.9 %)	1.44 ± 0.11 (7.4 %)	Ductile
[16]	Recycled PC and sand	1.86	65–73	51–71	1.43–2.44	Shear (brittle), ductile
[24]	Recycled HDPE	0.86	11.19	//	//	Shear (brittle)
[24]	Recycled PP	0.87	10.02	//	//	//
[49,50]	Concrete bricks	1.80	7	//	//	Brittle

testing orientations of the samples and the contribution of the metal particle orientation to crack propagation, with visible detached portions along the fracture surface of the Pb samples (Fig. 5b).

The results from the compression tests demonstrate the potential use of the MRE-based retentate to produce hot-pressing construction bricks. Although the use of pyrolysis byproducts as the main material for building elements is still scarcely explored, some considerations can be made by comparing the results of this work with data from bricks made of plastic-based waste composites or recycled plastics. According to the state-of-the-art, there is greater interest in using similar waste as fillers, reinforcements, or aggregates for concrete rather than for fully or almost fully plastic-based components, further highlighting the potential use of the MRE-based retentate. As shown in Table 2, bricks made of recycled polycarbonate (PC) and sand can achieve higher compressive properties and, by decreasing the sand percentage content, ductile fractures [16]. However, their density is comparable to conventional bricks [49,50], making them less suitable for lightweight applications. Recycled thermoplastics, such as HDPE and PP, can achieve densities comparable with the MRE-based retentate studied in this work, although their mechanical properties are less suitable for structural applications, e.g., lower compressive strength and brittle shear fracture behavior [24]. Overall, the MRE-based retentate represents a balance between compressive mechanical properties and low densities thanks to its heterogeneous composition, making it a potential alternative for applications requiring good strength-to-weight ratios in the building sector.

3.4. Applications and potential fields

According to the results, the tested composite material shows potential for the construction sector, especially for lightweight bricks and structures. First, its compressive properties match the standard specification for building bricks from ASTM C62-23 [52]. As shown in Table 2, the compressive strength values are well above the range of construction bricks and brickworks (~ 7 MPa) [49,53] with a significant decrease in density and, hence, weight [50]. This fact provides advantages of the embodied energy of transportation and ease of handling. Assuming an average density of 1.80 g/cm^3 , a standard concrete brick of $92 \times 57 \times 203 \text{ mm}$ dimensions weighs 1.92 Kg, while a pyrolysis waste composite brick would weigh 0.82 Kg, achieving a percentage reduction of 57.3 %, with significant savings in energy consumption and costs. A brick made with the tested material is also 12.5 % lighter than lightweight bricks, whose weight is 0.94 Kg [53,54], representing an alternative with better compressive properties. In addition, the ductile behavior of the MREs-based retentate can help predict their failure by providing visible signs of damage and early indications of failure for structural components, e.g., brickworks and columns, improving the safety of structures over conventional part or brick fractures. This behavior also simplifies part replacement before rupture compared to standard brick materials, making the material tested in this study a valuable option for the building sector, such as columns and indoor panels. Its density and strength-to-weight ratio make this material a good option to reduce the weight of building structures, taking advantage of the composite nature of the MRE-based retentate. Considering the potential waste volume from MREs, i.e., ~ 14117 tons per year [39], this material can be used as lightweight bricks for brickworks in negligible weathering, with promising uses for moderate weathering after tuning the material formulation [52], reducing the need for virgin materials. Thanks to the presence of PE, PET, and wax in the formulation, the composite might be used for applications connected to energy conservation, as well as improved water resistance and heat barrier properties [29,46,48] despite the need for further safety assessment, e.g., flammability. Lastly, the ability of this material to be manufactured through hot-pressing and its easy moldability suggest potential uses of the MRE-based retentate for applications with complex geometries and surface details, such as customized textures for interiors.

The MRE-based retentate could be suitable for different applications in the building sector, such as: (i) small batches of building parts with intricate geometries, e.g., façades and curved or textured panels; (ii) components with higher dimensional accuracy thanks to the limited shrinkages and omitted drying and firing steps, e.g., non-standard paving slabs, non-circular columns and pillars, or indoor dividers and screens; (iii) aesthetic architectural and decorative lightweight elements exploiting the texture of the composite formulation, e.g., decorative façades, sculptural elements and replicas, or indoor textured dividers; (v) indoor structures and products for interior design taking advantage of lightweight, ductile, compression-resistant, and heat barrier materials, e.g., pot, containers, and racks; (v) building elements with weight and safety constraints, including during transportation and handling, e.g., lightweight bricks or panels and partitions; or (vi) pavements, walkways, or driveway parts in novel composite formulations with sand or aggregates, e.g., customized textured slabs or cover tiles. In summary, this material from the industrial pyrolysis process represents an alternative to

reduce the need for virgin construction materials, detecting a range of potential structural and aesthetic applications for waste composites.

3.5. Limitations and future work

Despite the promising results, this work presents some limitations to be addressed in future work. First, the material needs a deeper characterization to fully assess its suitability for the proposed applications. Other properties should be quantified to provide a complete overview of the material. These characterizations can range from evaluating its flowability and processability with the selected manufacturing process, e.g., rheological tests and thermal characterization, to its durability and safety as part of building structures, such as fatigue, weathering, and flammability tests. Second, the work focused on a specific waste stream, which can be affected by the specific batch variability and material compositions. Tests can be done on retentate byproducts from different MRE waste, assessing the variability of the building material according to the different waste sources and improving the potential scalability of the results. In addition, the work mainly focused on the compressive properties of the new material, lacking in providing a comprehensive overview of the other mechanical properties, e.g., from tensile and flexural tests. Adding them would strengthen the rationale of the potential applications previously discussed. Finally, limitations can also be connected to the difficulties in standardizing the manufacturing process, which requires additional work to define the optimal processing parameters and protocols. Nevertheless, these limitations can be seen as potential future directions for exploiting this byproduct and similar materials in the construction sector.

Future work can also be done to deepen the characterization of this composite material and further expand the range of possible applications. In detail, future research should focus on (i) detecting potential application products and proofs-of-concept, such as construction elements for building and architecture; (ii) deepening the study of the hot compression molding process, defining the optimal processing parameters and their influence on the final parts; (iii) using further manufacturing processes and studying their processability, e.g., additive manufacturing for large-format components with intricate geometries [55]; (iv) deepening the characterization of this formulation, studying its thermal behavior, other mechanical properties, and the morphology of the matrix and dispersed phases of the composite material; and (v) studying specific material formulations, e.g., with sand or aggregates, deepening the characterization of its properties according to the selected sectors, e.g., durability, weather resistance, temperature ranges, and flame-retardancy modifications [25]. Finally, a (vi) life cycle assessment (LCA) should be performed to quantify the environmental impact of moving to this alternative construction material, e.g., GHG or energy demand. Similarly, a (vii) technoeconomic analysis (TEA) of the process to produce the pyrolysis waste composite brick can determine profitability at different scales of production, assessing its economic viability.

4. Conclusions

For the first time, filtered contaminants from liquid-fed pyrolysis were studied as alternative materials for the building sector, repurposing it as a fully recycled secondary raw material for building components and interior parts. The practical application of this work lies in repurposing industrial recycling byproducts from post-consumer multilayer packaging, the MRE-based retentate, as a novel composite material, taking advantage of its intrinsic heterogeneous composition and fostering resource retention principles. After pyrolysis and filtering, the MRE-based retentate composition was investigated to define its composition and suitability as a new secondary raw material for construction. Two batches of specimens were manufactured by hot compression molding to assess the manufacturability of the material and study its compression properties, allowing for comparisons within the building sector. The main results achieved in this work are summarized below:

- As shown from the TGA, the retentate collected as a byproduct from the liquid-fed pyrolysis of post-consumer multilayer food packaging is a thermoplastic-based composite, which is composed of pyrolysis wax and a PE/PET mixture as a matrix and aluminum particles (layer film) as the main dispersed phase. Despite the need for additional thermal characterization, this lightweight composition is well-suited to the building sector, combining mechanical performances with potential enhancements of the final parts in water resistance, heat barrier properties, and energy conservation.
- The samples were successfully fabricated by hot pressing, reaching a good overall quality in terms of material integrity and homogeneity. The ease of moldability and postprocessing demonstrated the possible use of hot pressing to manufacture construction elements and bricks, also removing drying and firing times, hence minimizing shrinkage and lowering energy consumption. At the same time, the results suggest the feasibility of more intricate parts and complex shapes, paving the way for architectural and decorative applications with aesthetic textures from the material, as well as the use of other manufacturing technologies, e.g., additive manufacturing.
- According to the mechanical tests, the compressive behavior of the material is superior to common building elements, such as construction bricks and brickworks, e.g., reaching ~10–12 MPa of compressive strength. An anisotropic behavior was observed in the samples, with higher compressive properties when the compression load is parallel to the mold opening, e.g., a 54 % increase in the compressive yield strength. The ductile fracture behavior can help predict the failure of parts made with this material, making them suitable for applications requiring higher safety for construction structures.
- The MRE-based retentate shows a significantly lower density, i.e., 0.77 g/cm³, compared to conventional materials for bricks. Its lightweight composition leads to considerable weight reductions in brick manufacturing, which means 57.3 % lighter than conventional clay bricks and 12.5 % lighter than lightweight bricks. This result can pave the way for a wider range of applications in

structures and pavement parts requiring a good strength-to-weight ratio, reducing overall weights, transportation energy, and costs.

- The unique property combination of the MRE-based retentate shows potential for several applications in the building sector, including (i) building parts with intricate geometries, e.g., customized facades and textured panels; (ii) components with limited shrinkage and good dimensional accuracy, e.g., non-circular columns and indoor dividers; (iii) aesthetic and decorative lightweight parts, e.g., sculptural elements and decorative facades; (iv) interior design products, e.g., pots and racks; (v) lightweight building components with predictable ductile failures, e.g., bricks or partitions; and (vi) pavement or driveway parts, e.g., textured slabs and tiles. Their use can foster resource retention, reducing virgin material and energy usage.

Although further characterization is required, this secondary raw material represents a promising alternative to reduce the need for virgin materials in the building sector, finding a meaningful application for composite materials from industrial byproducts. According to the ongoing shift towards circular practices, future research in this area should detect other discarded byproducts to be exploited for similar sectors, closing the loop and avoiding their disposal. Considering them as new composite materials can increase the range of available options on the market, achieving unique property combinations thanks to their intrinsic composite nature of mixed or contaminated byproducts, e.g., good strength-to-weight ratios. Finally, digital manufacturing technologies, such as additive manufacturing, should be further explored to exploit industrial byproducts in the building sector, customizing material properties and achieving more complex geometries.

CRedit authorship contribution statement

Alessia Romani: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Daniel Kulas:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation. **Joseph Curro:** Methodology, Investigation, Data curation. **David R. Shonnard:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Joshua M. Pearce:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

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Data availability

Data will be made available on request.

References

- [1] E. Elhacham, L. Ben-Uri, J. Grozovski, Y.M. Bar-On, R. Milo, Global human-made mass exceeds all living biomass, *Nature* 588 (7838) (Dec. 2020) 442–444, <https://doi.org/10.1038/s41586-020-3010-5>.
- [2] S. Zinatloo-Ajabshir, M.S. Morassaei, M. Salavati-Niasari, Eco-friendly synthesis of Nd₂Sn₂O₇-based nanostructure materials using grape juice as green fuel as photocatalyst for the degradation of erythrosine, *Compos. B Eng.* 167 (Jun. 2019) 643–653, <https://doi.org/10.1016/j.compositesb.2019.03.045>.
- [3] S. Zinatloo-Ajabshir, et al., Novel rod-like [Cu(phen)₂(OAc)]·PF₆ complex for high-performance visible-light-driven photocatalytic degradation of hazardous organic dyes: DFT approach, Hirshfeld and fingerprint plot analysis, *J. Environ. Manag.* 350 (Jan. 2024) 119545, <https://doi.org/10.1016/j.jenvman.2023.119545>.
- [4] H. Rahimpour, A.B. Amini, F. Sharifi, A. Fahmi, S. Zinatloo-Ajabshir, Facile fabrication of next-generation sustainable brick and mortar through geopolymerization of construction debris, *Sci. Rep.* 14 (1) (May 2024) 10914, <https://doi.org/10.1038/s41598-024-61688-x>.
- [5] H. Rahimpour, A. Fahmi, S. Zinatloo-Ajabshir, Toward sustainable soda ash production: a critical review on eco-impacts, modifications, and innovative approaches, *Results Eng.* 23 (Sep. 2024) 102399, <https://doi.org/10.1016/j.rineng.2024.102399>.
- [6] N. Mohamad, K. Muthusamy, R. Embong, A. Kusbiantoro, M.H. Hashim, Environmental impact of cement production and Solutions: a review, *Mater. Today Proc.* 48 (Jan. 2022) 741–746, <https://doi.org/10.1016/j.matpr.2021.02.212>.
- [7] B.C. McLellan, R.P. Williams, J. Lay, A. van Riessen, G.D. Corder, Costs and carbon emissions for geopolymer pastes in comparison to ordinary portland cement, *J. Clean. Prod.* 19 (9) (Jun. 2011) 1080–1090, <https://doi.org/10.1016/j.jclepro.2011.02.010>.
- [8] G.Z. Batista dos Santos, D. Passos de Oliveira, J. de Almeida Melo Filho, N. Marques da Silva, Sustainable geopolymer composite reinforced with sisal fiber: durability to wetting and drying cycles, *J. Build. Eng.* 43 (Nov. 2021) 102568, <https://doi.org/10.1016/j.jobbe.2021.102568>.

- [9] V. Mymrin, et al., Sustainable material manufacturing from hazardous bauxite red mud in composites with clay slate waste and lime production waste, *Int. J. Adv. Manuf. Technol.* 111 (5–6) (Nov. 2020) 1375–1385, <https://doi.org/10.1007/s00170-020-06187-9>.
- [10] C. Abdy, Y. Zhang, J. Wang, Y. Cheng, I. Artamendi, B. Allen, Investigation of high-density polyethylene pyrolyzed wax for asphalt binder modification: mechanism, thermal properties, and ageing performance, *J. Clean. Prod.* 405 (Jun. 2023) 136960, <https://doi.org/10.1016/j.jclepro.2023.136960>.
- [11] A. Fahmi, S.R. Zavaragh, M.R. Hanafi, H. Rahimpour, S. Zinatloo-Ajabshir, A. Asghari, Facile preparation, characterization, and investigation of mechanical strength of Starchy NaCl-binder as a lightweight construction material, *Sci. Rep.* 13 (1) (Nov. 2023) 19042, <https://doi.org/10.1038/s41598-023-46536-8>.
- [12] M.R. Hanafi, H. Rahimpour, S. Zinatloo-Ajabshir, F. Moodi, A. Fahmi, Performance enhancement, life cycle assessment, and feature analysis of wheat starch-based NaCl-binder as a sustainable alternative to OPC mortar, *Results Eng.* 24 (Dec. 2024) 103281, <https://doi.org/10.1016/j.rineng.2024.103281>.
- [13] H.T. Mohan, K. Jayanarayanan, K.M. Mini, Recent trends in utilization of plastics waste composites as construction materials, *Constr. Build. Mater.* 271 (Feb. 2021) 121520, <https://doi.org/10.1016/j.conbuildmat.2020.121520>.
- [14] T.K. Meyer, N.G. Tanikella, M.J. Reich, J.M. Pearce, Potential of distributed recycling from hybrid manufacturing of 3-D printing and injection molding of stamp sand and acrylonitrile styrene acrylate waste composite, *Sustain. Mater. Technol.* 25 (Sep. 2020) e00169, <https://doi.org/10.1016/j.susmat.2020.e00169>.
- [15] F.I. Aneke, C. Shabangu, Green-efficient masonry bricks produced from scrap plastic waste and foundry sand, *Case Stud. Constr. Mater.* 14 (Jun. 2021) e00515, <https://doi.org/10.1016/j.cscm.2021.e00515>.
- [16] M.C. Woods, A. Kulkarni, J.M. Pearce, The potential of replacing concrete with sand and recycled polycarbonate composites: compressive strength testing, *J. Compos. Sci.* 7 (6) (Jun. 2023), <https://doi.org/10.3390/jcs7060249>. Art. no. 6.
- [17] L. Dai, et al., Pyrolysis technology for plastic waste recycling: a state-of-the-art review, *Prog. Energy Combust. Sci.* 93 (Nov. 2022) 101021, <https://doi.org/10.1016/j.pecs.2022.101021>.
- [18] M. Kusenber, et al., Assessing the feasibility of chemical recycling via steam cracking of untreated plastic waste pyrolysis oils: feedstock impurities, product yields and coke formation, *Waste Manag.* 141 (Mar. 2022) 104–114, <https://doi.org/10.1016/j.wasman.2022.01.033>.
- [19] A. Zolghadr, D. Kulas, D. Shonnard, Evaluation of pyrolysis wax as a solvent in polyolefin pyrolysis processing, *Ind. Eng. Chem. Res.* 61 (30) (Aug. 2022) 11080–11088, <https://doi.org/10.1021/acs.iecr.2c01493>.
- [20] D.G. Kulas, A. Zolghadr, D.R. Shonnard, Liquid-fed waste plastic pyrolysis pilot plant: effect of reactor volume on product yields, *J. Anal. Appl. Pyrolysis* 166 (Sep. 2022) 105601, <https://doi.org/10.1016/j.jaap.2022.105601>.
- [21] M.S. Klippel, M.F. Martins, Physicochemical assessment of waxy products directly recovered from plastic waste pyrolysis: review and synthesis of characterization techniques, *Polym. Degrad. Stabil.* 204 (Oct. 2022) 110090, <https://doi.org/10.1016/j.polymdegradstab.2022.110090>.
- [22] M. Geissdoerfer, P. Savaget, N.M.P. Bocken, E.J. Hultink, The Circular Economy – a new sustainability paradigm? *J. Clean. Prod.* 143 (Feb. 2017) 757–768, <https://doi.org/10.1016/j.jclepro.2016.12.048>.
- [23] M.E. Grigore, Methods of recycling, properties and applications of recycled thermoplastic polymers, *Recycling* 2 (4) (Dec. 2017), <https://doi.org/10.3390/recycling2040024>. Art. no. 4.
- [24] P. Kulkarni, V. Ravekar, P. Rama Rao, S. Waigokar, S. Hingankar, Recycling of waste HDPE and PP plastic in preparation of plastic brick and its mechanical properties, *Cleaner Mater.* 5 (Sep. 2022) 100113, <https://doi.org/10.1016/j.clema.2022.100113>.
- [25] L. Gu, T. Ozbakkaloglu, Use of recycled plastics in concrete: a critical review, *Waste Manag.* 51 (May 2016) 19–42, <https://doi.org/10.1016/j.wasman.2016.03.005>.
- [26] M.Y. Khalid, Z.U. Arif, W. Ahmed, H. Arshad, Recent trends in recycling and reusing techniques of different plastic polymers and their composite materials, *Sustain. Mater. Technol.* 31 (2022), <https://doi.org/10.1016/j.susmat.2021.e00382>.
- [27] M. Saberian, et al., Recycling of spent coffee grounds in construction materials: a review, *J. Clean. Prod.* 289 (Mar. 2021) 125837, <https://doi.org/10.1016/j.jclepro.2021.125837>.
- [28] J.M. Maguire, V. Koutsos, D. Ray, Vacuum-infused thermoplastic fibre-metal laminates – advances in bonding and recycling, *Mater. Today Commun.* 37 (Dec. 2023) 106961, <https://doi.org/10.1016/j.mtcomm.2023.106961>.
- [29] M.A.R. Bhuiyan, et al., Heat insulating jute-reinforced recycled polyethylene and polypropylene bio-composites for energy conservation in buildings, *Mater. Today Commun.* 37 (Dec. 2023) 106948, <https://doi.org/10.1016/j.mtcomm.2023.106948>.
- [30] D. Ayre, Technology advancing polymers and polymer composites towards sustainability: a review, *Curr. Opin. Green Sustainable Chem.* 13 (2018) 108–112, <https://doi.org/10.1016/j.cogsc.2018.06.018>.
- [31] Global plastic waste generation by application 2019–2060, Statista. Accessed: October. 10, 2024. [Online]. Available: <https://www.statista.com/statistics/1339200/global-plastic-waste-generation-outlook-by-application/>.
- [32] D. Jin, T.K. Meyer, S. Chen, K. Ampadu Boateng, J.M. Pearce, Z. You, Evaluation of lab performance of stamp sand and acrylonitrile styrene acrylate waste composites without asphalt as road surface materials, *Constr. Build. Mater.* 338 (Jul. 2022) 127569, <https://doi.org/10.1016/j.conbuildmat.2022.127569>.
- [33] H. Li, J. Yang, D. Yang, N. Zhang, S. Nazar, L. Wang, Fiber-reinforced polymer waste in the construction industry: a review, *Environ. Chem. Lett.* 22 (6) (Dec. 2024) 2777–2844, <https://doi.org/10.1007/s10311-024-01769-5>.
- [34] M.C.S. Ribeiro, et al., Re-use assessment of thermoset composite wastes as aggregate and filler replacement for concrete-polymer composite materials: a case study regarding GFRP pultrusion wastes, *Resour. Conserv. Recycl.* 104 (Nov. 2015) 417–426, <https://doi.org/10.1016/j.resconrec.2013.10.001>.
- [35] E.C. Ailenei, et al., New waste-based composite material for construction applications, *Materials* 14 (20) (Jan. 2021), <https://doi.org/10.3390/ma14206079>. Art. no. 20.
- [36] R. Roy, M. Mottaghi, M. Woods, J.M. Pearce, Life cycle carbon emissions savings of replacing concrete with recycled polycarbonate and sand composite, *Sustainability* 17 (3) (2025) 839, <https://doi.org/10.3390/su17030839>.
- [37] Y. Wang, et al., Catalytic pyrolysis of polyethylene for the selective production of monocyclic aromatics over the zinc-loaded ZSM-5 catalyst, *ACS Omega* 7 (3) (Jan. 2022) 2752–2765, <https://doi.org/10.1021/acsomega.1c05401>.
- [38] C. Vargas, A.E. Hanandeh, The utilisation of polyethylene waste pyrolytic wax as an additive to enhance the performance of virgin and polypropylene-modified bitumen, *Int. J. Pavement Res. Technol.* 17 (4) (Jul. 2024) 929–951, <https://doi.org/10.1007/s42947-023-00278-y>.
- [39] J.A. Ratto, J. Lucciari, C. Thellen, D. Froio, N.A. D'Souza, Reduction of solid waste associated with military rations and packaging, Technical Report (2010) 11. <https://serdp-estcp.mil/projects/details/8b5f57b2-35a0-4330-a34b-41405fdbdfb9>, 2 [Online]. Available: (Accessed 17 October 2024).
- [40] F.K. Hafting, et al., Modular open-source design of pyrolysis reactor monitoring and control electronics, *Electronics* 12 (24) (Jan. 2023), <https://doi.org/10.3390/electronics12244893>. Art. no. 24.
- [41] Plastic shredder: recycling solution for plastics and 3D prints [3devo. <https://www.3devo.com/gp20-plastic-shredder>. (Accessed 9 October 2024).
- [42] M.C. Woods, C.K. Brooks, J.M. Pearce, Open-source cold and hot scientific sheet press for investigating polymer-based material properties, *HardwareX* 19 (Sep. 2024) e00566, <https://doi.org/10.1016/j.ohx.2024.e00566>.
- [43] M. Wang, M. Zhang, Z. Lu, W. Jia, Hot pressing technology investigation of light-weight wooden brick, *Constr. Build. Mater.* 265 (Dec. 2020) 120426, <https://doi.org/10.1016/j.conbuildmat.2020.120426>.
- [44] M. Idrees, A. Akbar, F. Saeed, M. Gull, S.M. Eldin, Sustainable production of Low-Shrinkage fired clay bricks by utilizing waste plastic dust, *Alex. Eng. J.* 68 (Apr. 2023) 405–416, <https://doi.org/10.1016/j.aej.2023.01.040>.
- [45] ASTM D695 – 23, Standard test method for compressive properties of rigid plastics. <https://compass.astm.org/document/?contentCode=ASTM%7CD0695-23%7Cen-US&proxycl=, 2023>. (Accessed 29 August 2024), 2Fsecure.astm.org&fromLogin=true.
- [46] A.A. Cuadri, C. Delgado-Sánchez, A. Tenorio-Alfonso, P. Partal, F.J. Navarro, Form-stable bitumen/paraffin-wax/polymer binders for energy-efficient building applications, *J. Energy Storage* 93 (Jul. 2024) 112420, <https://doi.org/10.1016/j.est.2024.112420>.
- [47] A. Kurdi, N. Almoatham, M. Mirza, T. Ballweg, B. Alkahlan, Potential phase change materials in building wall construction—a review, *Materials* 14 (18) (Jan. 2021), <https://doi.org/10.3390/ma14185328>. Art. no. 18.
- [48] N.F.A. Halim, N. Taib, Z.A. Aziz, The performance of thermal property in concrete containing waste pet (polyethylene terephthalate) as an alternative sustainable building material, *IOP Conf. Ser. Earth Environ. Sci.* 452 (1) (Apr. 2020) 012108, <https://doi.org/10.1088/1755-1315/452/1/012108>.

- [49] Compression and tension strength of some common materials. https://www.engineeringtoolbox.com/compression-tension-strength-d_1352.html. (Accessed 2 October 2024).
- [50] Brick densities. https://www.engineeringtoolbox.com/bricks-density-d_1777.html. (Accessed 10 October 2024).
- [51] Density of plastics material: technical properties table. <https://omnexus.specialchem.com/polymer-property/density>. (Accessed 10 October 2024).
- [52] ASTM C62-33, Standard specification for building brick (solid masonry units made from clay or shale). <https://compass.astm.org/document/?contentCode=ASTM%7CC0062-23%7Cen-US>. (Accessed 15 October 2024).
- [53] E. Escalera, G. Garcia, R. Terán, R. Tegman, M.-L. Antti, M. Odén, The production of porous brick material from diatomaceous earth and Brazil nut shell ash, *Constr. Build. Mater.* 98 (Nov. 2015) 257–264, <https://doi.org/10.1016/j.conbuildmat.2015.08.003>.
- [54] K. Pimraksa, P. Chindaprasirt, Lightweight bricks made of diatomaceous earth, lime and gypsum, *Ceram. Int.* 35 (1) (Jan. 2009) 471–478, <https://doi.org/10.1016/j.ceramint.2008.01.013>.
- [55] A. Romani, M. Levi, Large-format material extrusion additive manufacturing for circular economy practices: a focus on product applications with materials from recycled plastics and biomass waste, *Sustainability* 16 (18) (Jan. 2024), <https://doi.org/10.3390/su16187966>. Art. no. 18.